# Turoi

## Digital Pathology Economic Projection

*An evidence-based decision instrument*

| | |
|---|---|
| **INSTITUTION** | Standalone Diagnostic Lab |
| **REGION** | India (INR) |
| **PROJECTION HORIZON** | 5 years |
| **DATE PREPARED** | 2026-04-13 |
| **METHODOLOGY HASH** | 20289e83 |

*This projection is derived from institution-specific inputs combined with benchmark data from 20 published sources in peer-reviewed journals, government pay commissions, and sector analyst reports. All default assumptions carry citations, confidence levels, and source attributions. The full reference list in Vancouver format is appended at the end of this document.*

# Disclaimer & Limitations

This document has been generated by Turoi, an evidence-based decision-support instrument. The economic projections presented herein are derived from institution-specific user inputs combined with benchmark data from 20 published sources in peer-reviewed journals, government pay commission reports, and sector analyst research.

All default values carry explicit citations, confidence levels (high, medium, or low), and source attributions, visible in the Assumptions & Methodology appendix of this document and in the interactive methodology tab of the Turoi application. Where the AI analyst has adjusted values, the original value, the adjusted value, and the reasoning are recorded in the assumptions ledger.

This document does not constitute financial advice, a formal health economic evaluation (cost-effectiveness analysis, cost-utility analysis, or budget impact analysis), a health technology assessment submission, a regulatory filing, or a guarantee of financial outcomes. The net present value, return on investment, and payback projections are modelled estimates subject to the assumptions stated herein.

Uncertainty intervals (90% confidence bands) are generated via Monte Carlo simulation (2,000 samples) with triangular perturbation of each input scaled by its assigned confidence level. These intervals reflect modelling variation only. They do not capture exogenous factors including but not limited to: procurement timeline variability, regulatory or reimbursement policy changes, macroeconomic shifts, vendor pricing changes, staffing market conditions, or institutional restructuring.

All assumptions should be independently reviewed and validated by the institution's own financial, clinical, and operational leadership before any capital commitment or procurement decision is made.

The authors of Turoi accept no liability for decisions made on the basis of this projection.

Methodology hash: 20289e83. This hash uniquely identifies the assumption set used to generate this projection and can be used for audit and reproducibility purposes.

## Confidence Levels

Every assumption in the Turoi model is assigned a confidence level that reflects how well-supported it is by published evidence. This level directly affects the width of the Monte Carlo uncertainty band for that input:

| Level | Monte Carlo perturbation | Meaning |
|---|---|---|
| **High** | +/- 10% | Well-established from two or more independent published sources. Example: US OR cost per minute (Childers 2018 + Thomas 2022). |
| **Medium** | +/- 20% | Supported by one published source or extrapolated from closely related data. Example: Indian pathologist salary derived from the 7th Central Pay Commission pay matrix. |
| **Low** | +/- 35% | Rough estimate based on limited or indirect evidence, grey literature, or analyst inference. Example: data monetisation value per terabyte of curated WSI data. |

The 'Conf.' column in the Assumptions & Methodology appendix shows this level for every value. Inputs with low confidence produce wider uncertainty bands and are the first candidates for institutional validation.

# Executive Summary

The digital pathology business case for this standalone diagnostic lab presents a negative investment outcome, with a projected NPV of -₹39.61 lakh over the planning horizon. The 90% confidence interval ranges from indicating consistent negative returns across sensitivity scenarios. The investment fails to achieve payback within the projection period and delivers a -9.8% ROI.

Three primary value drivers partially offset the implementation costs but prove insufficient to justify the investment. Pathologist productivity improvements represent the strongest pillar at ₹40.17 lakh, likely reflecting integration and reduced microscopy time. Commercial growth through B2B partnerships c potential for expanded service offerings to hospital networks and reference laboratory relationships. Hub and spoke logistics optimization adds ₹24.65 lakh in value, indicating benefits from centralized digital revie transport costs.

The negative outcome reflects structural challenges inherent to standalone diagnostic labs in the Indian market. Limited reimbursement coverage, with heavy reliance on self-pay and restricted insurance acceptance, constrains revenue scaling potential. The lean single-site staffing model, while cost-efficient, may limit the operational leverage needed to maximize digital pathology's productivity benefits. Direct vendor procurement and owner-based decision-making suggest capital constraints that could impact implementation scope and quality.

The B2B reimbursement component offers the most promising growth vector, as corporate health screening contracts and hospital partnerships typically provide more predictable revenue streams than individual patient services. However, the projected commercial growth pillar alone cannot overcome the investment's fundamental economics.

For a department considering this technology transition, the analysis suggests that digital pathology implementation would require either significant modifications to the business model, expanded service catchment area, or improved reimbursement contract negotiations to achieve financial viability. The productivity gains and logistics efficiencies are meaningful but insufficient under current operational parameters.

The primary uncertainty lies in the commercial growth projection's achievability, as B2B market penetration rates and pricing sustainability in India's competitive diagnostic landscape remain difficult to predict with precision.

## Institution Profile

| | |
|---|---|
| **Region** | India (INR) |
| **Institution type** | Standalone Diagnostic Lab |
| **Attribution mode** | diagnostic lab |
| **Procurement model** | direct vendor + owner decision |
| **Staffing model** | lean_single_site |
| **Reimbursement** | self-pay / limited insurance / B2B |
| **Annual slide volume** | 10,000 |
| **Annual case volume** | 15,000 |
| **Pathologist FTE** | 2 |
| **Histotech FTE** | 4 |

# Key Metrics

At the current assumptions, the deployment does not recover its cost within 5 years. The largest contributor is pathologist productivity at Rs. 40.17 L, but total savings of Rs. 1.48 Cr do not exceed total cost of Rs. 1.64 Cr.

| | |
|---|---|
| **Net Present Value (NPV)** | **Rs. -39.61 L** |
| **90% Confidence Interval (Monte Carlo)** | **Rs. -57.62 L to Rs. -19.24 L** |
| **Payback Year** | **Beyond horizon** |
| **Return on Investment (ROI)** | **-9.8%** |
| **Cost per Case (over horizon)** | **Rs. 219** |
| **Productivity Hours Recaptured / year** | **720 hours** |
| **Total Capital Expenditure** | **Rs. 92.50 L** |
| **Operating Expenditure (Year 1)** | **Rs. 12.90 L** |

## Value Pillar Breakdown

The projection identifies 8 active value pillars. Each pillar represents a distinct category of economic impact attributable to the digital pathology deployment. Pillars are tagged by attribution level to indicate where in the institution the savings or revenue accrue.

| Value Pillar | Total (5 years) | Attribution |
|---|---|---|
| Pathologist productivity | **Rs. 40.17 L** | Lab Opex |
| Commercial growth (B2B) | **Rs. 35.99 L** | Commercial Growth |
| Hub & spoke logistics | **Rs. 24.65 L** | Lab Opex |
| Waste reduction & QA | **Rs. 16.54 L** | Lab Opex |
| Slide movement & supply chain | **Rs. 16.36 L** | Lab Opex |
| Risk & compliance | **Rs. 9.13 L** | Lab Opex |
| Subspecialty routing | **Rs. 3.56 L** | Commercial Growth |
| Consult revenue scalability | **Rs. 1.92 L** | Commercial Growth |

# Year-by-Year Projection

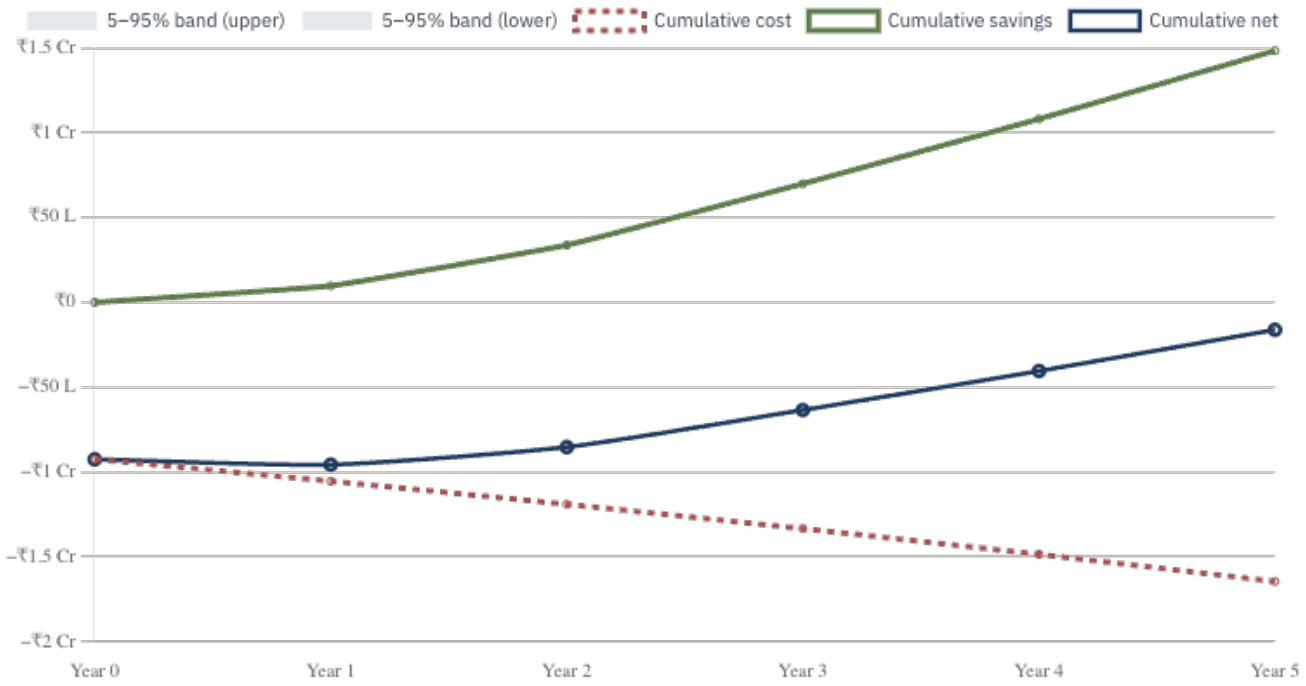
The table below shows the annual cash-flow projection for the 5-year horizon. Year 0 capital expenditure (Rs. 92.50 L) is incurred upfront. Operating expenditure begins in Year 1 at Rs. 12.90 L and is inflated annually. Savings grow according to the productivity ramp: 25% of steady-state in Year 1, 60% in Year 2, and 100% from Year 3 onward.

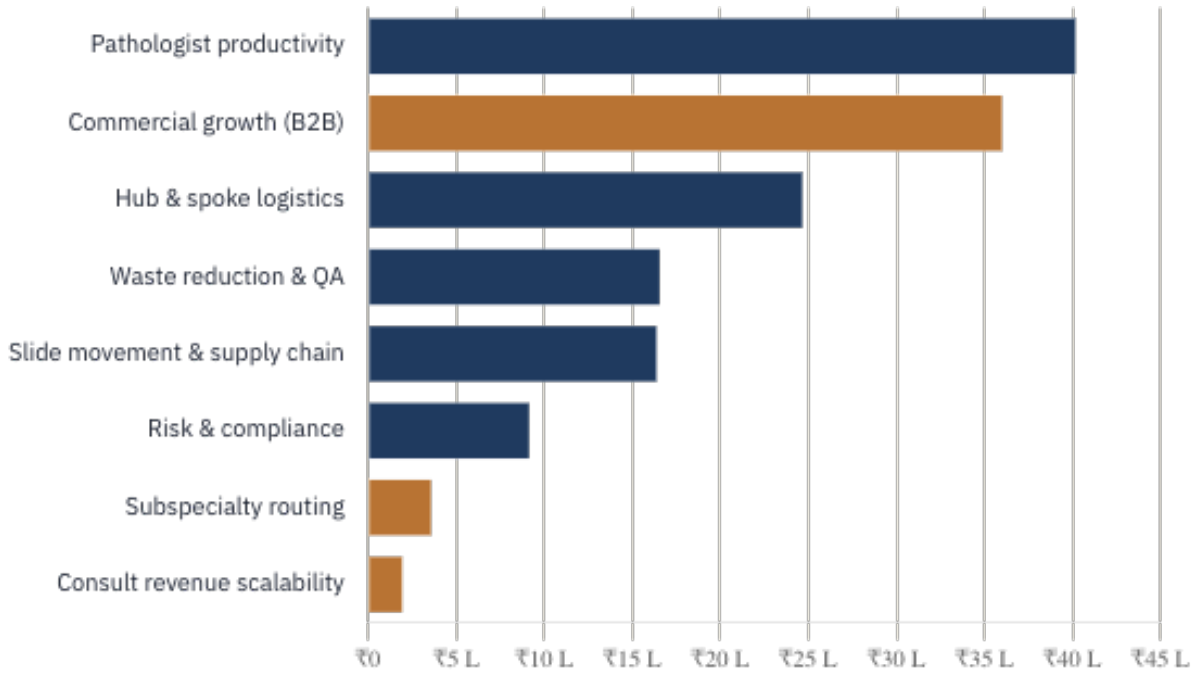
| Year | Investment | Savings | Net | Cumulative |
|---|---|---|---|---|
| Year 0 | Rs. 92.50 L | Rs. 0 | Rs. -92.50 L | **Rs. -92.50 L** |
| Year 1 | Rs. 12.90 L | Rs. 9.75 L | Rs. -3.15 L | **Rs. -95.65 L** |
| Year 2 | Rs. 13.61 L | Rs. 24.00 L | Rs. 10.39 L | **Rs. -85.26 L** |
| Year 3 | Rs. 14.36 L | Rs. 36.17 L | Rs. 21.81 L | **Rs. -63.46 L** |
| Year 4 | Rs. 15.15 L | Rs. 38.16 L | Rs. 23.01 L | **Rs. -40.45 L** |
| Year 5 | Rs. 15.98 L | Rs. 40.25 L | Rs. 24.27 L | **Rs. -16.17 L** |

## Pillar Detail by Year

| Pillar | Y0 | Y1 | Y2 | Y3 | Y4 | Y5 | Total |
|---|---|---|---|---|---|---|---|
| **Slide movement & supply chain** | Rs. 0 | Rs. 1.08 L | Rs. 2.65 L | Rs. 3.99 L | Rs. 4.21 L | Rs. 4.44 L | **Rs. 16.36 L** |
| **Pathologist productivity** | Rs. 0 | Rs. 2.64 L | Rs. 6.50 L | Rs. 9.79 L | Rs. 10.33 L | Rs. 10.90 L | **Rs. 40.17 L** |
| **Waste reduction & QA** | Rs. 0 | Rs. 1.09 L | Rs. 2.68 L | Rs. 4.03 L | Rs. 4.26 L | Rs. 4.49 L | **Rs. 16.54 L** |
| **Risk & compliance** | Rs. 0 | Rs. 60.0k | Rs. 1.48 L | Rs. 2.23 L | Rs. 2.35 L | Rs. 2.48 L | **Rs. 9.13 L** |
| **Hub & spoke logistics** | Rs. 0 | Rs. 1.62 L | Rs. 3.99 L | Rs. 6.01 L | Rs. 6.34 L | Rs. 6.69 L | **Rs. 24.65 L** |
| **Subspecialty routing** | Rs. 0 | Rs. 23.4k | Rs. 57.6k | Rs. 86.8k | Rs. 91.6k | Rs. 96.6k | **Rs. 3.56 L** |
| **Commercial growth (B2B)** | Rs. 0 | Rs. 2.37 L | Rs. 5.82 L | Rs. 8.78 L | Rs. 9.26 L | Rs. 9.77 L | **Rs. 35.99 L** |
| **Consult revenue scalability** | Rs. 0 | Rs. 12.6k | Rs. 31.0k | Rs. 46.7k | Rs. 49.3k | Rs. 52.0k | **Rs. 1.92 L** |

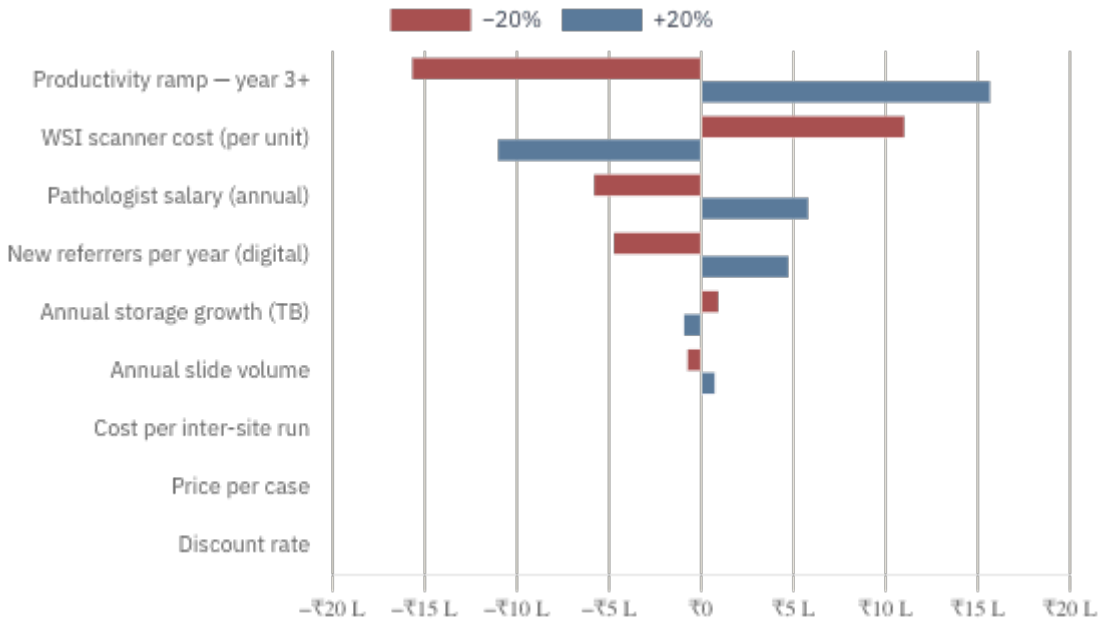
# Financial Projection Charts



# Sensitivity Analysis

Each key input is varied by +/-20% from its base value while holding all other inputs constant. The resulting change in NPV is reported below. The most sensitive input is productivity ramp — year 3+, where a 20% change produces an NPV swing of Rs. 15.65 L. Inputs with large magnitude should be validated with the highest priority, as errors in these assumptions have the greatest impact on the projected outcome.




## Sensitivity Table (each input varied +/- 20%)

| Input | NPV at +20% | NPV at -20% | Magnitude |
|---|---|---|---|
| Productivity ramp — year 3+ | Rs. 15.65 L | Rs. -15.65 L | **Rs. 15.65 L** |
| WSI scanner cost (per unit) | Rs. -11.00 L | Rs. 11.00 L | **Rs. 11.00 L** |
| Pathologist salary (annual) | Rs. 5.79 L | Rs. -5.79 L | **Rs. 5.79 L** |
| New referrers per year (digital) | Rs. 4.72 L | Rs. -4.72 L | **Rs. 4.72 L** |
| Annual storage growth (TB) | Rs. -92.1k | Rs. 92.1k | **Rs. 92.1k** |
| Annual slide volume | Rs. 72.7k | Rs. -72.7k | **Rs. 72.7k** |
| Cost per inter-site run | Rs. 0 | Rs. 0 | **Rs. 0** |
| Price per case | Rs. 0 | Rs. 0 | **Rs. 0** |
| Discount rate | Rs. 0 | Rs. 0 | **Rs. 0** |

# Assumptions & Methodology

The table below lists every numeric assumption used in this projection together with its source, evidence basis, confidence level, and citation reference. Confidence levels (high, medium, low) indicate how well-established each value is in published literature and directly determine the width of the Monte Carlo uncertainty perturbation applied to that input. Values entered by the user or adjusted by the AI analyst are flagged separately.

| CAPEX | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| WSI scanner cost (per unit) | Rs. 55.00 L | user | user-entered in wizard | medium | |
| Number of scanners | 1.00 | user | user-entered in wizard | medium | |
| Reporting workstation cost | Rs. 1.80 L | region default | | medium | |
| Number of workstations | 3.00 | ai_adjusted | Adjusted by AI analyst during pre-estimation review | medium | |
| Implementation services cost | Rs. 22.00 L | region default | | medium | |
| Validation / go-live cost | Rs. 8.00 L | region default | | medium | |
| Training cost per FTE | Rs. 35.0k | region default | | medium | |

| CLINICAL OPS | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| Annual slide volume | 10,000.00 | user | Single-site standalone labs typically handle 30–60k slides/yr; larger volumes... | medium | [18] |
| Annual case volume | 15,000.00 | user | user-entered in wizard | medium | |

| COMMERCIAL GROWTH | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| B2B referring physicians (current) | 35.00 | ai_adjusted | Adjusted by AI analyst during pre-estimation review | medium | |
| Avg volume per referrer | 320.00 | region default | | medium | |
| New referrers per year (baseline) | 2.00 | ai_adjusted | Adjusted by AI analyst during pre-estimation review | medium | |
| New referrers per year (digital) | 8.00 | region default | | medium | |
| Referrer retention uplift from digital | 8.0% | region default | | medium | |
| Margin per case | Rs. 280 | CRISIL Indian diagnostics margin study | Standalone lab gross margin per histopathology case after consumables and lab... | medium | [18] |
| Price per case | Rs. 650 | region default | | medium | |
| TAT reputation factor | 4.0% | region default | | medium | |

| CONSULT REVENUE SCALE | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| External consult price per case | Rs. 800 | user | user-entered in wizard | medium | |
| Projected consult volume growth | 35.0% | region default | | medium | |

| CONSULTS | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| Consults sent (annual) | 100.00 | user | user-entered in wizard | medium | |
| Consults received (annual) | 150.00 | user | user-entered in wizard | medium | |
| Consult courier cost per case | Rs. 250 | user | user-entered in wizard | medium | |
| Current consult TAT (days) | 10.00 | Vodovnik 2018 and Indian B2B referral TAT data | Physical slide transport to subspecialty consult is dominant delay; 7–14 days... | medium | [4] |
| Projected digital consult TAT (days) | 1.00 | region default | | medium | |

| HUB SPOKE | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| Total sites | 1.00 | region default | | medium | |
| Spoke collection centers | 0.00 | region default | | medium | |
| Inter-site courier runs per week | 0.00 | region default | | medium | |
| Cost per inter-site run | Rs. 0 | region default | | medium | |
| Inter-site sample loss / damage rate | 0.3% | region default | | medium | |

| INFRASTRUCTURE | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| Archive space (sqft) | 500.00 | ai_adjusted | Adjusted by AI analyst during pre-estimation review | medium | |
| Archive rent per sqft per year | Rs. 420 | region default | | medium | |

| OPEX | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| IMS software license (annual) | Rs. 9.00 L | user | user-entered in wizard | medium | |
| IMS support & AMC (annual) | Rs. 2.20 L | region default | | medium | |
| Cloud egress & data transfer (annual) | Rs. 60.0k | region default | | medium | |
| Storage cost per TB per year | Rs. 11.0k | region default | | medium | |
| Annual storage growth (TB) | 10.00 | region default | | medium | |

| PROJECTION | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| Discount rate | 10.0% | region default | | medium | |
| Projection horizon (years) | 5.00 | region default | | medium | |
| Inflation rate | 5.5% | region default | | medium | |
| Productivity ramp — year 1 | 30.0% | Vodovnik 2024 adjusted for smaller site | Smaller labs tend to realize productivity faster due to shorter validation cy... | medium | [5] |

| PROJECTION | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| Productivity ramp — year 2 | 70.0% | region default | | medium | |
| Productivity ramp — year 3+ | 100.0% | region default | | medium | |

| RISK | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| Medicolegal requests per year | 12.00 | region default | | medium | |
| Cost saved per legal request | Rs. 20.0k | region default | | medium | |

| SLIDE MOVEMENT | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| Courier cost (annual) | Rs. 4.50 L | region default | | medium | |
| Slide retrieval hours per week | 12.00 | region default | | medium | |

| STAFFING | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| Pathologist FTE | 2.00 | ai_adjusted | Adjusted by AI analyst during pre-estimation review | medium | |
| Histotechnologist FTE | 4.00 | ai_adjusted | Adjusted by AI analyst during pre-estimation review | medium | |
| Pathologist salary (annual) | Rs. 22.00 L | Standalone lab pay benchmarks | Owner-operator or 1–3 staff pathologist model; salary reflects market-rate pr... | medium | |
| Histotechnologist salary (annual) | Rs. 4.20 L | region default | | medium | |

| STRATEGIC | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| Consult revenue per case | 0.00 | user | User-provided value | high | |

| SUBSPECIALTY ROUTING | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| External consults per year | 400.00 | user | user-entered in wizard | medium | |
| External consult cost per case | Rs. 800 | user | Out-of-network second-opinion fees charged by national chains and academic ce... | medium | |
| Internal consult cost per case | Rs. 500 | user | user-entered in wizard | medium | |
| Subspecialty out-referral rate | 12.0% | region default | | medium | |
| Subspecialty retained with digital | 65.0% | region default | | medium | |

| WASTE | Value | Source | Evidence basis | Confidence | Ref |
|---|---|---|---|---|---|
| IHC stain cost per slide | Rs. 400 | user | user-entered in wizard | medium | |
| IHC stains avoided per year | 900.00 | region default | | medium | |
| Recut cost per slide | Rs. 100 | user | user-entered in wizard | medium | |
| Recuts avoided per year | 650.00 | region default | | medium | |
| Lost slide remediation cost | Rs. 12.0k | region default | | medium | |
| Lost slides avoided per year | 18.00 | region default | | medium | |

# References

*All default values, benchmark coefficients, and regional calibrations are derived from the following published sources.*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